

Mar 18 01:37

Spyder (Python 3.11)

File Edit Search Source Run Debug Consoles Projects Tools Window Help

...me/yusuf/Desktop/AI_ML_Progs/Mobile_price_prediction_project

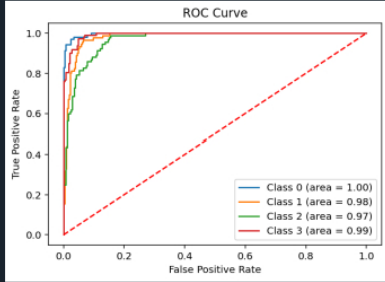
/home/yusuf/Desktop/AI_ML_Progs/Mobile_price_prediction_project/mobile_price_prediction.py

mobile_price_prediction.py x

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 from sklearn.model_selection import train_test_split
7 from sklearn.ensemble import RandomForestClassifier
8 from sklearn.preprocessing import StandardScaler
9 from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, roc_curve, auc
10
11 print("Loading Dataset...")
12
13 data = pd.read_csv("train.csv")
14
15 print("\nFirst 5 rows:")
16 print(data.head())
17
18 print("\nDataset Shape:", data.shape)
19
20 print("\nMissing Values:")
21 print(data.isnull().sum())
22
23 plt.figure(figsize=(12,10))
24 sns.heatmap(data.corr(), cmap="coolwarm")
25 plt.title("Feature Correlation Heatmap")
26 plt.show()
27
28 X = data.drop("price_range", axis=1)
29 y = data["price_range"]
30
31 scaler = StandardScaler()
32 X_scaled = scaler.fit_transform(X)
33
34 X_train, X_test, y_train, y_test = train_test_split(
35     X_scaled, y, test_size=0.2, random_state=42
36 )
37
38 print("\nTraining Samples:", len(X_train))
39 print("Testing Samples:", len(X_test))
40
41 print("\nTraining Model...")
42
43 model = RandomForestClassifier(
44     n_estimators=500,
45     max_depth=15,
46     random_state=42
47 )
```

44 %

ROC Curve



Class 0 (area = 1.00)
Class 1 (area = 0.98)
Class 2 (area = 0.97)
Class 3 (area = 0.99)

Console 1/A x

0.0 0.2 0.4 0.6 0.8 1.0

False Positive Rate

Enter mobile features to predict price range:
Battery Power: 1200
RAM: 512
Pixel Height: 480
Pixel Width: 800
Mobile Weight: 180
Talk Time: 8

Predicted Price Range: 0
Low Cost Phone

In [2]:

IPython Console History

Inline Conda: ai_classes (Python 3.11.14) LSP: Python Line 14, Col 1 UTF-8 LF RW Mem 74%

Mar 18 01:38

Spyder (Python 3.11)

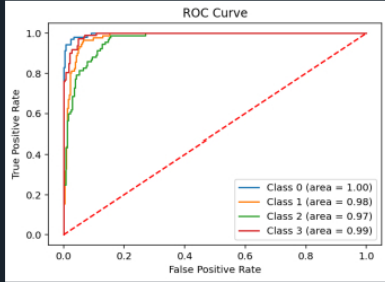
File Edit Search Source Run Debug Consoles Projects Tools Window Help

...me/yusuf/Desktop/AI_ML_Progs/Mobile_price_prediction_project

mobile_price_prediction.py

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 from sklearn.model_selection import train_test_split
7 from sklearn.ensemble import RandomForestClassifier
8 from sklearn.preprocessing import StandardScaler
9 from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, roc_curve, auc
10
11 print("Loading Dataset...")
12
13 data = pd.read_csv("train.csv")
14
15 print("\nFirst 5 rows:")
16 print(data.head())
17
18 print("\nDataset Shape:", data.shape)
19
20 print("\nMissing Values:")
21 print(data.isnull().sum())
22
23 plt.figure(figsize=(12,10))
24 sns.heatmap(data.corr(), cmap="coolwarm")
25 plt.title("Feature Correlation Heatmap")
26 plt.show()
27
28 X = data.drop("price_range", axis=1)
29 y = data["price_range"]
30
31 scaler = StandardScaler()
32 X_scaled = scaler.fit_transform(X)
33
34 X_train, X_test, y_train, y_test = train_test_split(
35     X_scaled, y, test_size=0.2, random_state=42
36 )
37
38 print("\nTraining Samples:", len(X_train))
39 print("Testing Samples:", len(X_test))
40
41 print("\nTraining Model...")
42
43 model = RandomForestClassifier(
44     n_estimators=500,
45     max_depth=15,
46     random_state=42
47 )
```

ROC Curve



Class 0 (area = 1.00)
Class 1 (area = 0.98)
Class 2 (area = 0.97)
Class 3 (area = 0.99)

Console 1/A

```
Enter mobile features to predict price range:
Battery Power: 4500
RAM: 8192
Pixel Height: 1440
Pixel Width: 3040
Mobile Weight: 165
Talk Time: 25

Predicted Price Range: 3
Very High Cost Phone

In [3]:
```

IPython Console History

Inline Conda: ai_classes (Python 3.11.14) LSP: Python Line 14, Col 1 UTF-8 LF RW Mem 78%